

1 Solution — GIAM 1.7

Problem 2

1.1 Problem

The *domain* of a function (or binary relation) is the set of numbers appearing in the first coordinate; the *range* is the set of numbers appearing in the second coordinate. Consider the set $\{0, 1, 2, 3, 4, 5, 6\}$ and the function $f(x) = x^2 \pmod{7}$. Express this function as a relation by explicitly writing out the set of ordered pairs it contains. What is the range of this function?

1.2 Answer

The function as a set of ordered pairs $(x, f(x))$:

$$f = \{ (0, 0), (1, 1), (2, 4), (3, 2), (4, 2), (5, 4), (6, 1) \}$$

and its range — the set of second coordinates — is

$$\text{range}(f) = \{0, 1, 2, 4\}.$$

1.3 The Computation

Square each element of the domain and reduce modulo 7 (i.e. take the remainder after dividing by 7):

x	x^2	$x^2 \pmod{7}$	pair
0	0	0	(0, 0)
1	1	1	(1, 1)
2	4	4	(2, 4)
3	9	2	(3, 2)

4	16	2	(4, 2)
5	25	4	(5, 4)
6	36	1	(6, 1)

Table 1.1

The **domain** is all seven first coordinates $\{0, 1, 2, 3, 4, 5, 6\}$; the **range** is the set of distinct second coordinates, $\{0, 1, 2, 4\}$ — only four values, because several inputs share an output.

1.4 The Function as a Mapping

$f(x) = x^2 \pmod{7}$. Several inputs collapse to the same output — so the range is smaller than the domain.

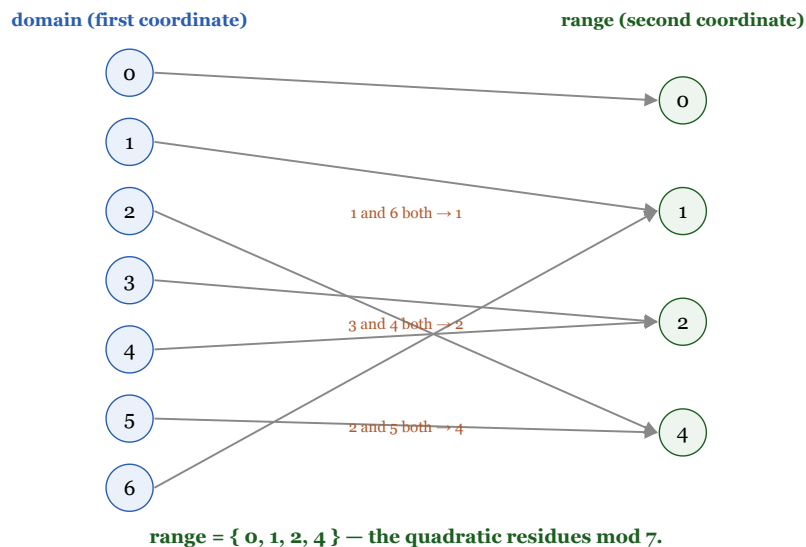


Figure 1.1: An arrow diagram. On the left, the domain nodes 0,1,2,3,4,5,6; on the right, the range nodes 0,1,2,4. Arrows: $0 \rightarrow 0$, $1 \rightarrow 1$, $6 \rightarrow 1$, $2 \rightarrow 4$, $5 \rightarrow 4$, $3 \rightarrow 2$, $4 \rightarrow 2$. The pairs 1 and 6 both map to 1; 2 and 5 both map to 4; 3 and 4 both map to 2; only 0 maps alone (to 0). So the range $\{0,1,2,4\}$ is the set of quadratic residues mod 7.

1.5 Remark — Why the Range Is Smaller Than the Domain (Non-Injectivity)

Seven inputs produce only four outputs, so f is **not injective** (not one-to-one): distinct inputs can share an output. The collapses are not random — they come from the symmetry $x^2 \equiv (7-x)^2 \pmod{7}$, since $(7-x)^2 = 49 - 14x + x^2 \equiv x^2$. So the inputs pair up as

$$1 \leftrightarrow 6, \quad 2 \leftrightarrow 5, \quad 3 \leftrightarrow 4,$$

each pair landing on a common value (1, 4, 2 respectively), while 0 is alone (it is its own “partner,” $7-0 \equiv 0$). That is exactly 3 pairs + 1 singleton = 4 outputs from 7 inputs.

1.6 Remark — The Range Is the Set of Quadratic Residues mod 7

The nonzero outputs $\{1, 2, 4\}$ are precisely the **quadratic residues** modulo 7 — the nonzero values that are perfect squares in modular arithmetic. For an odd prime p there are always exactly $\frac{p-1}{2}$ nonzero quadratic residues; here $\frac{7-1}{2} = 3$, matching $\{1, 2, 4\}$. The non-residues are the missing $\{3, 5, 6\}$ — no square is ever congruent to 3, 5, or 6 mod 7. Including 0, the full range $\{0, 1, 2, 4\}$ has $\frac{p-1}{2} + 1 = 4$ elements. This “roughly half of everything is a square” phenomenon is a cornerstone of number theory (it underlies quadratic reciprocity and much of modern cryptography).

1.7 Remark — Function-as-Relation, and the Vertical-Line Test

Writing f out as the pair-set $\{(0, 0), (1, 1), \dots\}$ is exactly the “function is a special relation” viewpoint the section describes. What makes this relation a **function** is that each domain element appears as a first coordinate *exactly once* — no input has two different outputs. That is the set-theoretic form of the vertical-line test: reading first coordinates as x and second as y , no two pairs share an x . Note the contrast with the *range*: outputs may repeat freely (and here they do), but inputs may not. A relation like $\{(2, 4), (2, 5)\}$ would fail — the input 2 would have two outputs — so it would not be a function.

1.8 Remark — Domain, Range, and Codomain Are Different Things

It is worth separating three ideas the problem’s vocabulary can blur. The **domain** $\{0, 1, \dots, 6\}$ is where inputs come from; the **range** (or image) $\{0, 1, 2, 4\}$ is the set of outputs actually achieved; and the **codomain** — the set outputs are *allowed* to fall in — here would naturally be all residues $\{0, 1, 2, 3, 4, 5, 6\}$. The function is **onto** (surjective) only if range = codomain, which fails here since $3, 5, 6$ are never hit. So f is neither injective nor surjective as a map $\{0, \dots, 6\} \rightarrow \{0, \dots, 6\}$. GIAM’s definition of range as “the numbers in the second coordinate” is precisely the *image* — the outputs that genuinely occur — which is why it can be a strict subset of the possible values.